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Designing artificial intelligence chatbots for self-regulated learning from a systematic review based on Habermas's three interests

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ABSTRACT

Self-regulated learning (SRL) is widely acknowledged as a key driver of academic success and lifelong learning, yet learners often struggle to plan, monitor, and reflect effectively on their progress. Concurrently, advances in artificial intelligence (AI) chatbots offer unprecedented opportunities to scale SRL. This paper reviews 78 studies from 2018 to 2025 on how AI chatbots can be created to support self-regulated learning (SRL) by leveraging technology, social interaction, and reflection, based on Habermas's three types of knowledge interests (technical, practical, and emancipatory). We identify eight core chatbot functions that align with the forethought, performance, and self-reflection phases of Zimmerman's SRL cycle. Socially, effective implementations embed chatbots within teacher–peer–chatbot triads to facilitate co-regulation, dialogue, and community building. Critically, chatbots can prompt deep self-evaluation and ethical inquiry, though designs must guard against overreliance and superficial reflection. We synthesise these findings into an integrative framework that maps chatbot features onto SRL processes and Habermasian interests, and we highlight underexplored areas – such as K-12 contexts, multimodal analytics, and emancipatory reflection – for future research. Our review offers actionable guidance for researchers, designers, and educators seeking to harness AI chatbots as versatile partners in fostering autonomous, socially grounded, and critically reflective learners.

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Artificial intelligence; Self-regulated learning; social co-regulation; metacognitive scaffolding; Habermas's knowledge-constitutive interests; chatbots

1. Introduction

Self-regulated learning (SRL) has long been recognised as a cornerstone of effective and lifelong learning, enabling students to set goals, monitor progress, and reflect upon outcomes in an autonomous, iterative cycle (Dignath & Veenman, 2021; Lawson et al., 2019; Zimmerman, 2002). In today's rapidly evolving educational landscape, learners must navigate complex informational environments and sustain their motivation – hallmarks of strong SRL competencies (de Bruin & van Merriënboer, 2017; Seufert, 2018; Wu & Chiu, 2025). Recent advancements in artificial intelligence (AI) have led to the development of sophisticated chatbots capable of delivering interactive and personalised learning support at scale, impacting how learners self-regulate their learning.

Recent developments in natural language processing (NLP) and large language models (e.g. GPT-3/GPT-4) have dramatically expanded chatbots' conversational fluency, enabling them to provide real-time feedback, tailor content to individual needs, and simulate dialogic exchanges once confined to human tutors (Kooli & Yusuf, 2025). Early research on educational AI chatbots predominantly emphasised task automation, such as administrative support or basic question-answer exchanges, thus addressing primarily surface-level learner needs. However, recent studies emphasise the opportunity for AI chatbots to support more sophisticated educational functions, including fostering learners' deeper cognitive and regulatory processes (Ng et al.,

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2024; Tsai et al., 2025; Wu & Chiu, 2025). To fully tap into these advanced capabilities, it is crucial to systematically explore how AI chatbots can (1) provide students with technological affordances, including adaptive feedback and personalised learning pathways; (2) facilitate meaningful social interactions, such as peer collaboration and co-regulatory dialogue; and (3) prompt reflective practices, thereby encouraging critical self-evaluation and transformative learning. Collectively, these three domains encompass critical dimensions of SRL, linking technological innovation with cognitive, social, and metacognitive theories of learning. These domains align with Habermas's three knowledge-constitutive interests: technical (technological affordances), practical (social interactions), and emancipatory (reflective practices).

Traditional SRL frameworks are excellent at simulating the internal cognitive and metacognitive processes of the individual learner. However, they frequently regard social and critical dimensions as contextual factors rather than central, constitutive elements of regulation. Habermas's (2015) framework of three knowledge-constitutive interests – technical, practical, and emancipatory – offers a multidimensional perspective on the ways in which AI chatbots can assist SRL in three ways. The framework (i) explicitly integrates the socio-technical nature of AI tools, (ii) offers a normative critical dimension absent in most SRL models, and (iii) serves as an integrative meta-framework.

This paper uses Habermas's three knowledge-constitutive interests as a framework to synthesise evidence on how AI chatbots can be engineered to foster SRL from instrumental, communicative, and critical perspectives (Ewert, 1991; Habermas, 2015). We adopt a systematic review methodology to map the landscape of empirical studies published between 2015 and early 2025, examining (1) the specific chatbot features that support goal setting, monitoring, and reflection; (2) the social configurations – teacher–chatbot, peer–chatbot, and co-regulatory networks – that underlie effective SRL; and (3) the ways in which chatbots can prompt critical self-evaluation and transformative reflection.

By weaving together evidence on technological affordances, social interaction, and reflective practice, this review advances both theory and practice. It ends with a combined framework that (a) connects specific chatbot features to Zimmerman's SRL phases and Habermas's knowledge-constitutive interests, (b) brings together common research methods, and (c) highlights important areas that need more investigation. This summary gives designers, educators, and researchers clear, research-based guidelines for using AI chatbots effectively to help develop independent, socially aware, and thoughtful learners.

2. Theoretical framework

2.1 Theoretical foundation of SRL

SRL refers to learners' active, iterative engagement in planning, monitoring, and evaluating their learning processes (Elhusseini et al., 2022; Pintrich & De Groot, 1990). Zimmerman's (2002) three-phase model is the most commonly used because it combines thinking, self-awareness, motivation, and actions into a clear process: *planning ahead* (setting goals, making plans, and analysing tasks), *doing the work* (controlling oneself and checking progress), and *looking back* (evaluating results, understanding causes, and making adjustments). This tripartite structure underpins many SRL measurement instruments and intervention studies, providing a practical scaffold for instructional design. Winne and Hadwin's (1998) COPES model extends Zimmerman's work by detailing how learners coordinate conditions, operations, products, evaluations, and standards within each regulatory episode. In parallel, metacognition theory clarifies how awareness of – and control over – cognitive strategies drives effective learning (Schraw & Dennison, 1994).

The frameworks reviewed above leave several important questions unresolved. First, the use of emerging digital agents, specifically AI chatbots, for SRL remains unclear. AI chatbots, for instance, engage in real-time interactions with learners' cognitive, motivational, and social processes. Second, most empirical SRL work still centres on individual cognition, offering only limited insight into the social or critical-reflective dimensions foregrounded by sociocultural and emancipatory perspectives. Third, there is limited theoretical development of how different layers of regulation (individual, co-regulated, and socially shared) are coordinated when an intelligent conversational agent becomes part of the learning system. Finally, the ethical and transformative implications of delegating regulatory functions to AI-driven tools remain unclear. Overall, these gaps signal the need for an integrative lens that can accommodate the *technical, social-communicative,* and *emancipatory* (Habermas's three interests) facets of SRL in AI-based environments.

2.2 Integrating Habermas' three interests and SRL with AI chatbots

While traditional SRL frameworks excel at modelling the internal cognitive and metacognitive processes of individual learners, they often treat social and critical dimensions as contextual factors rather than core, constitutive elements of regulation itself (e.g. Zimmerman's cyclical model, Winne & Hadwin's COPES model). In contrast, Habermas's framework (2015) of three knowledge-constitutive interests – *technical, practical,* and *emancipatory* – provides a multidimensional lens for analysing how AI chatbots can support SRL in three aspects.

First This framework could explicitly integrate the socio-technical nature of AI tools. It allows us to analyse not just that a chatbot scaffolds learning (technical interest), but how it structures social interaction and co-regulation (practical interest), and whether it fosters critical agency or induces passive dependency (emancipatory interest). The *technical* interest, oriented towards prediction, control, and efficiency, resonates with the cognitive–metacognitive dimension of SRL: chatbots operationalise this interest by personalising forethought and performance processes through adaptive feedback, automated strategy prompts, and data-driven analytics that optimise goal setting, strategic planning, and real-time self-monitoring. The *practical* (communicative) interest, which prioritises mutual understanding and social coordination, foregrounds the social-co-regulatory dimension of SRL; here, chatbots mediate dialogue among learners, peers, and teachers, scaffold peer feedback and teacher-student exchanges, and cultivate social presence, thereby extending regulation beyond the individual to shared and socially negotiated processes. The *emancipatory* interest, which focuses on helping people become more independent by challenging common beliefs, connects with the critical-reflective part of SRL: well-designed chatbots can encourage learners to examine their thinking, question the trustworthiness or bias of AI suggestions, and contemplate the ethical issues of using algorithms, thus promoting deep self-reflection and learner agency.

Second, this framework (2015) offers a normative critical dimension absent in most SRL models in three aspects. It provides a philosophical foundation to evaluate the quality and purpose of SRL support, moving beyond efficiency metrics to ask questions about power, autonomy, and transformative potential in human-AI interaction.

Third, this framework serves as an integrative meta-framework: it does not replace existing SRL models but rather subsumes them, positioning cognitive and metacognitive processes (the focus of Zimmerman's model) within the technical interest while adding the crucial practical and emancipatory dimensions necessary for a holistic analysis of learning in socially mediated, AI-augmented environments.

This comparative analysis makes a compelling case that the Habermasian lens is not merely an alternative organising scheme but a necessary theoretical advancement for understanding the multifaceted roles of AI in education. It specifically offers greater explanatory power for capturing the interplay among instrumental efficiency, social communication, and critical self-awareness that define next-generation learning with conversational agents, i.e. AI chatbots.

2.3 Technological affordances of AI chatbots for SRL

AI chatbots can be considered as multifaceted learning companions whose technological affordances map systematically onto the three phases of Zimmerman's SRL cycle (Chen et al., 2025; Chiu, 2024; Chiu et al., 2024; Labadze et al., 2023; Liu, Chiu, et al., 2024; Wang, 2024; Wu et al., 2024). First, at the forethought stage, chatbots optimise goal setting and strategic planning through adaptive task sequencing and natural language-driven content personalisation. By mining real-time learner inputs, they generate individualised learning trajectories and "just-in-time" hints that calibrate challenge and support (Chen et al., 2025; Labadze et al., 2023; Liu, Chiu, et al., 2024). Second, at the performance stage, feedback and metacognitive prompts can support learning. Immediate diagnostic feedback corrects errors and clarifies misconceptions, while embedded metacognitive prompts nudge learners to monitor their understanding (Shin et al., 2025). They can help learners complete the tasks needed during SRL (Lin & Chang, 2023; Liu, Chiu, et al., 2024). Third, during the self-reflection phase, chatbots afford automated analytics and formative assessment dashboards that visualise progress, compare current performance with prior attempts, and prompt learners to articulate revisions or alternative strategies (Labadze et al.,

2023). Collectively, the technological affordances of chatbots could enhance learners' SRL (i.e. their technical interests).

2.4 Social dimensions of chatbot-mediated SRL

SRL is fundamentally relational: learners plan, monitor, and reflect more effectively when those processes are embedded in supportive social networks. AI chatbots can be viewed as social catalysts that integrate teacher guidance, peer collaboration, and community engagement to support SRL. First, chatbots can facilitate teacher–student co-regulation. Teachers examine and verify AI-generated suggestions for SRL (Hew et al., 2023; Ng et al., 2024). This workflow creates a dialogic space to enable scaffolding across the forethought, performance, and reflection phases of SRL. Second, they can foster peer-to-peer co-regulation. They can allow learners to do peer activities such as role-play debates, peer commenting, and reviewing (Chen, 2024). Third, establishing a social presence at the community level is crucial. Chatbots cultivate a sense of belonging in online and blended courses (Labadze et al., 2023). Personalised greetings, adaptive group-formation algorithms, and chatbot-facilitated breakout discussions diminish isolation and sustain motivation (Labadze et al., 2023). Overall, chatbots have the potential to create supportive social environments for SRL (i.e. learners' practical interest).

2.5 Reflective and metacognitive practice with AI chatbots for SRL

Reflective practice – the capacity to evaluate one's own learning strategies and outcomes – constitutes the critical component of the self-reflection phase in Zimmerman's SRL cycle. AI chatbots can catalyse this phase by structuring iterative dialogue loops. Learners can compose or perform an initial response; the chatbot analyses that output, returns targeted prompts or exemplars, and invites the learner to revise or justify their choices. Each cycle externalises metacognitive questioning ("Why did I select this strategy?" "How does this revision address the feedback?") and gradually cultivates a finer-grained awareness of task requirements and personal heuristics (Lin & Chang, 2020).

Yet chatbot affordances alone do not guarantee deep reflection. Instructional mediation – whether by teachers, peer coaches, or instructional designers – operates along a continuum from minimal oversight to intensive guidance. When educators interrogate chatbot logs, visualise process data, and orchestrate guided debriefs, they help students move beyond surface corrections towards analytical commentary that identifies causal mechanisms and future transfer possibilities (Bai et al., 2025; Kim et al., 2025). Conversely, well-designed chatbot scaffolds can prompt critical questioning even in low-supervision settings, demonstrating that human mediation enhances but is not an absolute precondition for meaningful reflection.

Despite encouraging evidence that chatbots can prompt revision-orientated dialogue and surface learners' metacognitive reasoning, the reflective strand of the literature remains conceptually and empirically underdeveloped. For starters, studies rarely specify which component of reflection is being supported – evaluation, causal attribution, or strategy re-planning – or how these components align with established metacognitive constructs. Likewise, researchers seldom examine contextual moderators – for example, whether reflection unfolds differently in procedural disciplines (e.g. mathematics) versus ill-structured domains (e.g. writing) or at different educational levels. Lastly, very few studies connect reflection prompted by chatbots to positive results, like increased awareness of AI bias or more control for learners in choosing and evaluating algorithmic advice. These gaps highlight the need for a more systematic and robust synthesis of design features, measurement approaches, and contextual variables that shape the quality of chatbot-mediated metacognitive and critical-reflective practice.

3. The present study

3.1 Research gap and research questions

Prior reviews of AI chatbots in SRL have catalogued useful findings, but they remain conceptually fragmented, largely because they analyse single facets of SRL in isolation. This study uses Habermas's knowledge-constitutive interests – *technical, practical,* and *emancipatory* – as an organising lens to analyse the selected

articles. This unique lens is particularly advantageous for bringing together the research because it directly relates to (a) thinking and self-thinking, (b) social support and teamwork, and (c) critical thinking that we found in our previous review of self-regulated learning (SRL). As we discussed in the previous section, the four interrelated gaps are

- Technological dominance. Most syntheses foreground technical interest issues – for example, adaptive feedback and analytics – while offering scant insight into how chatbots structure the social processes of co-regulation that embody practical interest.
- Under-theorised reflection. Studies on reflective or journaling features typically treat them in isolation, without positioning them within an emancipatory, agency-orientated account of SRL.
- Lack of a comprehensive framework. No study to date aligns chatbot functionalities, social configurations, and reflective practices within a single explanatory model that spans all three Habermasian interests.
- Fragmented methodological landscape. Evidence is dispersed across disparate research designs and data sources, making it difficult to trace how methodological choices influence – or constrain – our theoretical understanding.

Therefore, we propose the following four research questions (RQs) to address the gaps:

RQ1: How can learning environments be designed using AI chatbots to foster SRL through technological affordances?

RQ2: In what ways do AI chatbots facilitate social interaction that supports the SRL process?

RQ3: How do AI chatbots promote critical self-reflection within SRL, enabling learners to examine and transform their learning strategies and underlying assumptions?

RQ4: How do AI chatbots support SRL when examined through the lens of Habermas's three knowledge interests?

3.2 Initial search and screen

On 28 February 2025, we searched both Web of Science and Scopus. After removing duplicate records, we conducted a two-stage screening process. First, titles and abstracts were reviewed against the pre-specified inclusion and exclusion criteria (section 3.3); second, full texts of all potentially relevant studies were obtained and assessed in detail for eligibility. Studies meeting all criteria were retained for data extraction (see Figure 1). The complete search strategy is provided in Table 1.

3.3 Selection criteria

We applied a set of predefined inclusion and exclusion criteria to ensure methodological rigour and direct relevance to AI chatbot-supported SRL, summarised in Table 2.

3.4 Quality appraisal

The methodological quality of all 78 studies was independently evaluated by two researchers using the criteria outlined by Theelen et al. (2019), which integrate standards for qualitative research (Savin-Baden & Major, 2007) and quantitative research (Health & Excellence, 2012) (Table 3). Each study was classified as qualitative (n = 9), quantitative (n = 28), or mixed-methods (n = 41), with mixed-methods articles appraised

Table 1. Search string

Dimension	Relevant keywords
AI	("AI" OR "artificial intelligence" OR "chatbot" OR "intelligent tutoring system" OR "conversational agent" OR "virtual agent" OR "virtual assistant" OR "autonomous learning agent" OR "automated tutor" OR "smart learning assistant" OR "digital tutor")
AND	
SRL	("SRL" OR "self-regulat*" OR "metacognition" OR "goal setting" OR planning OR "self-monitoring" OR "self-reflect*" OR "self-evaluat*" OR "motivation" OR autonomy OR "learning strateg*" OR "self-directed learning" OR "self-control*" OR "behavior regulat*" OR "emotion* regulat*" OR "self-efficacy" OR "resource management" OR "self-observ*")

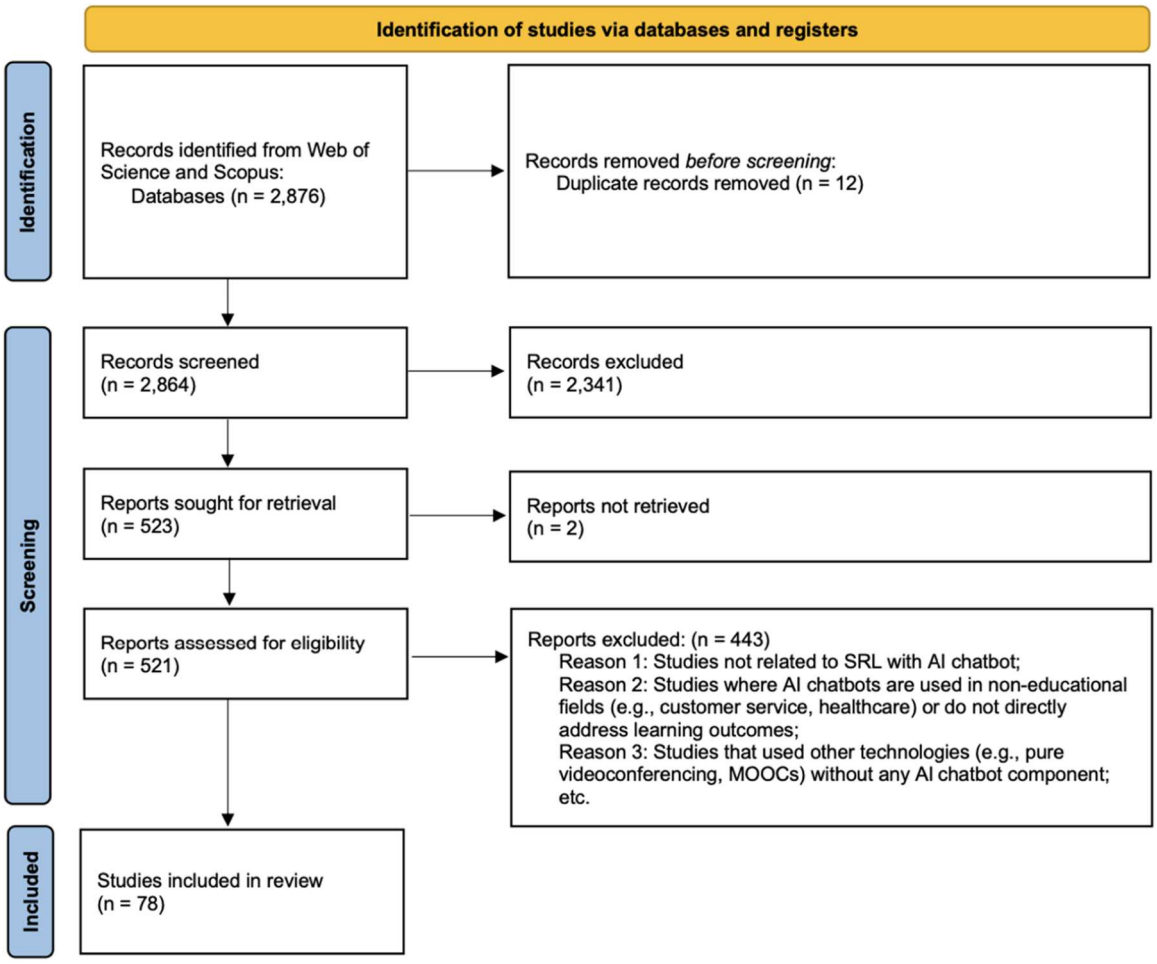


Figure 1. The selection process.

Table 2. Inclusion and exclusion criteria.

Inclusion	Exclusion
• Journal papers • Peer-reviewed • Written in English • Clear outcomes related to SRL (e.g. learner engagement, motivation, autonomy, learning performance, critical thinking, etc.) • Focus on AI Chatbots (e.g. rule-based, voice assistant) in education. • Teacher and Student Interaction • Relevance to SRL Frameworks	• Written in languages other than English • Editorials • Not peer-reviewed • Not relevant to AI chatbots and SRL • General AI Applications Without SRL Focus • Published before 2015 • Non-educational Contexts: Studies where AI chatbots are used in non-educational fields (e.g. customer service, healthcare) or do not directly address learning outcomes.

against both qualitative and quantitative benchmarks. Every criterion was rated on a four-point scale from 0 (no mention) to 3 (extensive mention). Inter-rater reliability was strong (Cohen's $\kappa = 0.84$), and any discrepancies were resolved through discussion until full consensus was reached. At this stage, we excluded none of the studies that achieved an average score above 2.0 ("Good mention"), leaving 78 articles for final analysis.

3.5 Data extraction and analysis

The following data were extracted from the included studies: bibliographic information (author, year, country), study design and sample characteristics, chatbot type and underlying technology, research methods, social interaction contexts, theoretical frameworks, and data collection methods.

For quantitative synthesis, we computed descriptive statistics to summarise study designs, data collection methods, sample size, subjects, and educational level (e.g. means and standard deviations). Guided by the a priori coding scheme described in Section 3.7, the qualitative analysis yielded themes that clarify how

Table 3. Quality appraisal criteria.

	0 No mention	1 Some mention	2 Good mention	3 Extensive mention
Criteria for qualitative studies				
Study is methodologically clear			5	45
Study is theoretically substantiated			3	47
Ethical process is transparent			10	40
Researcher(s) relation to participants is clear			17	33
Researcher relation to the data is clear		1	15	34
Researcher(s) take a critical stance towards own research			19	31
Methodology and methods used for data collection, analysis, and interpretation are congruent			1	49
Participant involvement in data interpretation		3	31	16
Limitations voiced			8	42
Criteria for quantitative studies				
Is the source population or source area well described?			4	65
Were interventions and comparisons well described and appropriate?			4	65
Were outcome measures reliable?			15	54
Were outcomes relevant			1	68
Were the analytical methods appropriate?			6	63
Are the study results internally valid (i.e. unbiased)?			37	32
Are the findings generalisable to the source population (i.e. externally valid)?		2	55	12

AI chatbots (a) scaffold the forethought, performance, and self-reflection phases of SRL, (b) facilitate teacher–peer–learner interactions, and (c) promote metacognitive and critical-reflective practices.

3.6 Coding scheme and data analysis

Following the hybrid content-analysis procedure of Elo and Kyngäs (2008), we combined deductive and inductive techniques. In the deductive phase, we developed an a priori coding frame aligned with our three analytic dimensions – *technological affordances, social interaction, and reflective practices* – grounded in SRL theory and Habermas's knowledge-constitutive interests. This frame was iteratively refined during the pilot coding of ten randomly selected studies. In the subsequent inductive phase, conceptually similar excerpts were consolidated into higher-order thematic categories. Code frequencies were then converted into descriptive statistics and visualised, while emergent themes served as analytic scaffolding for the narrative synthesis that addresses our RQs.

4. Results

4.1 Descriptive statistics

Appendix A provides an overview of each included study, listing author(s), publication year, educational level, discipline, SRL dimensions targeted, AI chatbot platform, research topics, research methods, sample size, and data collection techniques. The studies span publications from 2018 through 2025.

Figure 2 presents the predominant research methods, revealing that mixed-methods research (53%) was most common, followed by quantitative studies (36%) and qualitative research (12%). Figure 3 illustrates the distribution of sample sizes among the included studies, indicating that most studies involved medium-sized samples ranging from 30 to 200 participants (58.4%), while large samples exceeding 200 participants (22.1%) and smaller studies with fewer than 30 participants (19.5%) were less frequent.

Higher education settings conducted the vast majority of the included studies, predominantly involving undergraduate or graduate students (Figure 4). Secondary-school contexts constituted the next most frequent educational level, while research focused exclusively on primary education or on pre-service teacher training was comparatively rare. A small number of studies spanned K-12 through higher-education contexts, and several investigations targeted adult learners in professional-development or asynchronous online environments.

Language education, specifically ESL/EFL and Computer-Assisted Language Learning (CALL) contexts, constituted the majority of the included studies (Figure 5). A substantial portion examined STEM disciplines, including mathematics, physics, biology, and computer science/programming, while several adopted a

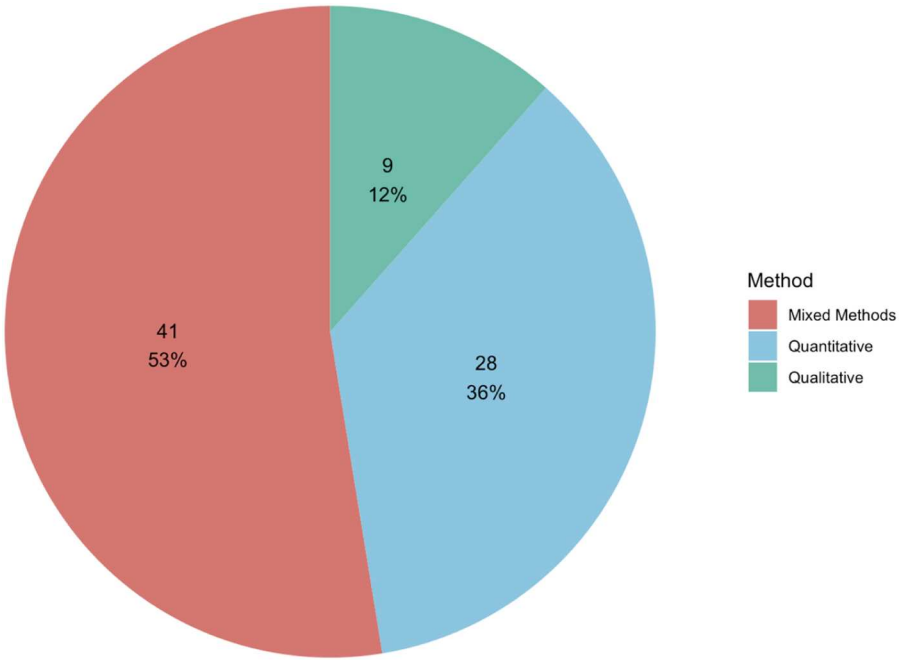


Figure 2. Distribution of research methods.

multidisciplinary lens – integrating digital storytelling, instructional design, or iSTEM (integrated STEM) approaches. A smaller but notable subset focused on health and life sciences (nursing, public health, food sciences, medical education), social sciences and education (psychology, teacher training, adult learning), and other fields such as business studies, music education, and digital game development for healthcare.

Most studies leveraged ChatGPT as their conversational engine, reflecting its rapid uptake in educational research. Beyond ChatGPT, prominent platforms included MetaTutor and Poe, while a smaller number employed specialised systems such as ELSA, Argumate, POObot, and PengTalk. Several investigations

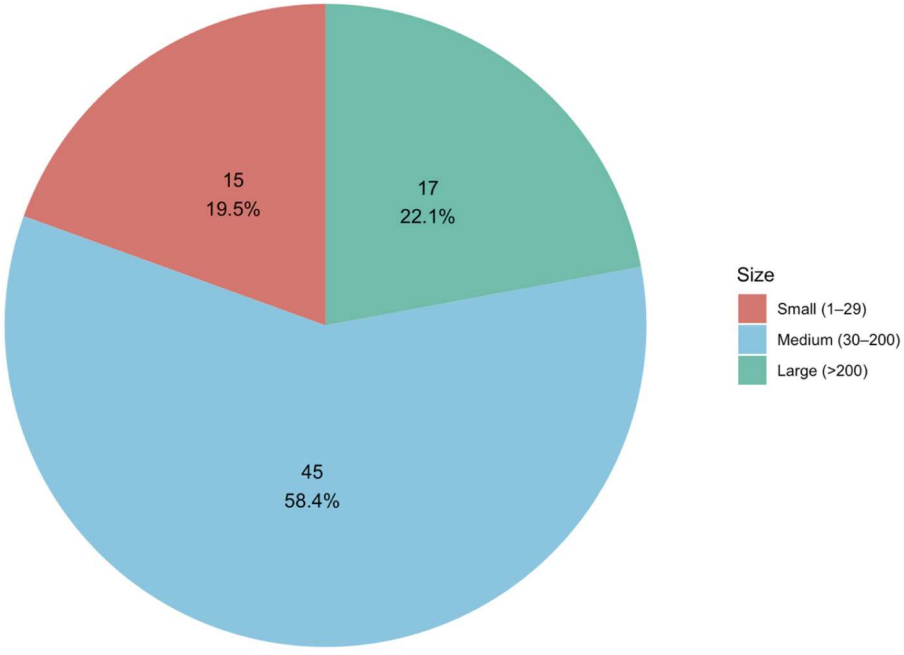


Figure 3. Distribution of sample size.

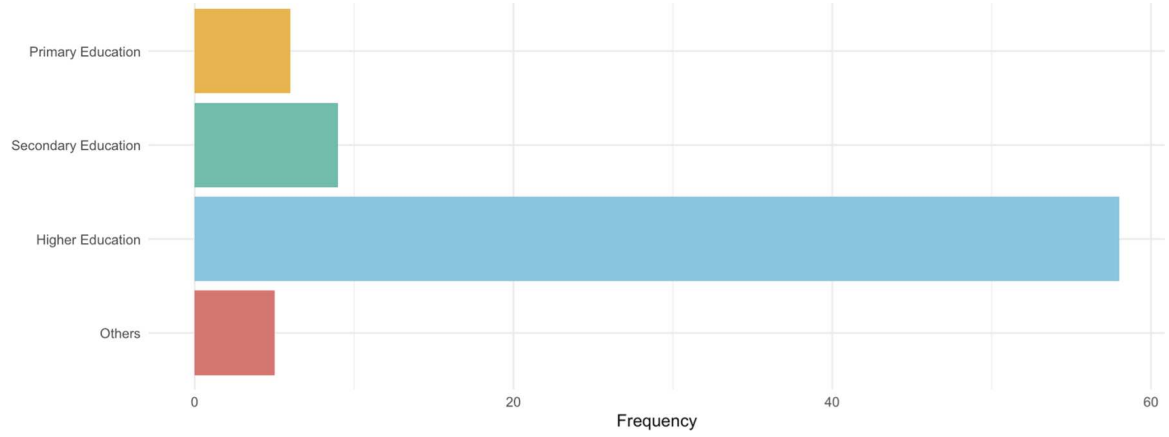


Figure 4. Educational levels distribution of included studies. Note: studies may appear in more than one category.

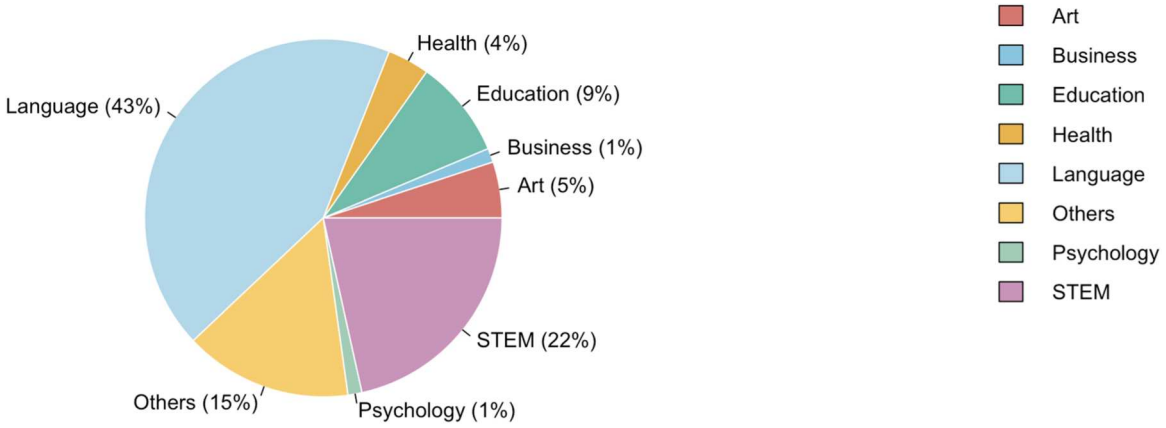


Figure 5. Distribution of studies by disciplines. Note: Studies that did not specify a single discipline or that spanned multiple disciplines have been grouped under "Others".

integrated chatbots into existing ecosystems – e.g. Telegram-based bots (BotFather/MenuBuilderBot), Slack via Flow XO, Dialogflow ("Study Buddy"), EDUguia, EXAIT, and Flow.ai – and a handful explored voice-activated assistants (Alexa Echo, Google Nest, and Microsoft's Xiao Bing).

Tables 4–6 present the coding outcomes mapped to RQ 1-3. For every component, the tables specify the categories and subcategories, offer operational definitions with illustrative excerpts from the literature, and cite all pertinent studies. Because a single study can populate more than one subcategory, the reported frequency counts are non-exclusive.

4.2 RQ1: how can learning environments be designed using AI chatbots to foster SRL through technological affordances?

Our synthesis revealed five principal categories of technological affordances through which AI chatbots scaffold SRL (Table 4). Chatbots personalise learning by dynamically adjusting task difficulty and sequencing based on learner inputs (Adams et al., 2024; Bahari et al., 2025); offering real-time, contextualised feedback on performance or writing (Abdelhalim, 2024); and enabling multimodal, scenario-based interactions (using text, voice, and visual media) to engage learners in problem-solving and digital storytelling (Al-Hafdi & Alnajdi, 2024).

Several studies integrated analytics to tailor learning trajectories and predict outcomes, using chatbot-generated data to recommend personalised paths (Huang et al., 2023) and to monitor task completion, engagement, and strategy use via log-file analysis (Jin et al., 2023; Kaiss et al., 2024).

Table 4. Technological affordances of AI chatbots for SRL.

Category	Subcategory	Specific functionality examples	References
Learning Design	Personalisation of Learning	• Adaptive learning paths (e.g. task design that adjusts based on learner progress) • Real-time, personalised feedback through AI chatbots (e.g. ChatGPT) • Content customisation to suit learner needs (e.g. GPT-3/4 adapting content based on user inputs)	(Adams et al., 2024) (Bahari et al., 2025) (Chang & Sun, 2025) (Chiu, 2024) (Duong & Chen, 2025) (Guo & Li, 2024) (Han et al., 2022) (Holster, 2024) (Kumar, 2021) (Lee et al., 2022) (Lee, 2024) (Levine et al., 2025) (Li, 2023) (Li, Sadiq, et al., 2025) (Li, Wang, et al., 2024) (Liang et al., 2024) (Liu, Hwang, et al., 2024) (Mills et al., 2025) (Naz & Robertson, 2024) (Ng et al., 2024) (Rahimi & Mosalli, 2025) (Robillos, 2024) (Saritepeci & Durak, 2024) (Shahzad et al., 2025)
	Feedback Mechanisms	• Immediate and contextualised feedback on performance, writing, or language tasks. • Adaptive feedback and automated assessment systems to support SRL processes (e.g. goal setting, performance monitoring).	(Abdelhalim, 2024) (Bai et al., 2025) (Chen et al., 2020) (Darvishi et al., 2024) (Fan et al., 2025) (Fan et al., 2025) (Guo et al., 2023) (Huang, 2024) (Kumar, 2021) (Levine et al., 2025) (Li, 2023) (Li, Sadiq, et al., 2025) (Liang et al., 2024) (Liu, Hwang, et al., 2024) (Mills et al., 2025) (Naz & Robertson, 2024) (Robillos, 2024) (Teng, 2025) (Zare et al., 2025)
	Interactive and Adaptive Learning Environments	• Use of NLP to facilitate dynamic, conversation-based interactions with learners. • Multimodal feedback (text, voice, and images) to engage learners in varied ways. • Use of chatbots for scenario-based learning (e.g. digital storytelling tasks, problem-solving exercises).	(Al-Hafdi & Alnajdi, 2024) (Chen & Law, 2025) (Guo et al., 2025) (Han et al., 2023) (Lee et al., 2022) (Li, 2023) (Li, Wang, et al., 2024) (Liu, Hwang, et al., 2024) (Liu & Reinders, 2025) (Mills et al., 2025) (Ng et al., 2024) (Nguyen & Nguyen, 2024) (Rahimi & Mosalli, 2025) (Rezai et al., 2024) (Saritepeci & Durak, 2024) (Tai & Chen, 2023) (Wang, Lin, et al., 2025) (Xiao et al., 2024) (Xie et al., 2024)
Learning Analytics	Data-Driven Personalisation	• Integration of learning analytics to track progress and predict learning outcomes. • Personalised learning paths based on predictive data and learner behaviour analysis (e.g. ChatGPT analytics on student progress).	(Huang et al., 2023) (Nguyen & Nguyen, 2024) (Teng et al., 2025) (Wang, Zhong, et al., 2024)
	Task and Performance Monitoring	• AI chatbots provide ongoing monitoring of task completion, learner engagement, and behaviour patterns. • Use of performance tracking to guide students through iterative learning processes (e.g. checking comprehension, revising drafts).	Jin et al. (2023); Kaiss et al. (2024); Khan-Galaria & Cukurova (2022); Pan et al. (2025); Wu et al. (2024);
Assessment	Automated Assessment and Scoring	• Use of AI-powered tools for real-time assessments and personalised scoring (e.g. essay evaluation using GPT models). • Integration of automated formative assessments in language learning, supporting SRL tasks like writing, reflection, and planning.	(Darmawansah et al., 2025) (Sayed et al., 2024)
	Goal-Setting and Reflection Support	• AI chatbots as tools for helping students set and revise learning goals. • AI as a tool for supporting **self-reflection** by prompting learners to evaluate their learning strategies and outcomes (e.g. iterative planning and evaluation).	(Cai et al., 2025) (Gazulla et al., 2023) (Han et al., 2025) (Harley et al., 2018) (Huang & Lin, 2024) (Kong et al., 2024) (Lee et al., 2022) (Levine et al., 2025) (Nguyen & Nguyen, 2024) (Robillos, 2024) (Saritepeci & Durak, 2024)
Learner Autonomy and Engagement	Support for Autonomy	• Chatbot features that empower learners to control their learning pace, task choices, and feedback. • AI chatbots encourage independent problem-solving through guided, scaffolded interactions.	(Durak, 2023) (Jiang et al., 2025) (Koc-Januchta et al., 2020) (Kong & Yang, 2024) (Kong et al., 2024) (Kumar, 2021) (Levine et al., 2025) (Rahimi & Mosalli, 2025) (Rezai et al., 2024) (Tram et al., 2024) (Wang, 2024) (Xia, Chiu, and Chai, 2023) (Xia, Chiu, Chai, and Xie, 2023) (Yin et al., 2024)
	Enhanced Learner Engagement	• Interactive and conversational features that maintain learner interest and motivation (e.g. gamified interactions, real-time collaborative support). • Use of AI for personalised motivation strategies (e.g. providing encouraging feedback, adaptive challenge levels).	(Jeon, 2022) (Kaiss et al., 2024) (Khallel, 2024) (Koc-Januchta et al., 2020) (Kong & Yang, 2024)
Collaborative Learning	Facilitating Collaborative Learning	• AI chatbots support group-based tasks by providing collaborative feedback, sharing resources, and guiding discussions. • Tools for collaborative note-taking, peer feedback, and	(Chen, 2024) (Khallel, 2024) (Levine et al., 2025)

(*Continued*)

Table 4. Continued.

Category	Subcategory	Specific functionality examples	References
		co-designing learning projects (e.g. Replika in collaborative note-taking systems).	
	Social Presence and Interaction	• AI chatbots enhance social presence in online environments, which helps reduce isolation in distance learning settings. • Enhancing peer engagement through chatbot-facilitated interactions, enabling joint attention and task coordination.	(Huang & Hew, 2025) (Khallel, 2024) (Kim et al., 2025)

Automated scoring engines provided immediate, personalised evaluation of essays and quizzes (Darmawansah et al., 2025), while built-in goal-setting prompts and reflective cues guided students through iterative planning and self-evaluation cycles (Cai et al., 2025; Gazulla et al., 2023).

Chatbots allowed learners to set their own speed, choose tasks, and ask for help whenever they needed it, which helped them become more independent and aware of their learning (Biyiri et al., 2025; Durak, 2023). They also used game-like chat features to keep learners motivated and engaged (Jeon, 2022; Kaiss et al., 2024).

By mediating peer and teacher interactions, chatbots facilitated group tasks – providing shared prompts, resources, and feedback – and enhanced social presence in online settings, reducing isolation and promoting co-regulation (Chen, 2024; Huang & Hew, 2025).

4.3 RQ2: in what ways do AI chatbots facilitate social interaction that supports the SRL process?

Our synthesis identified six interrelated ways in which AI chatbots mediate social interaction to support SRL (Table 5). Chatbots function as co-mediators alongside instructors, who design structured activities, verify chatbot outputs, and lead training sessions to ensure effective use (Abdelhalim, 2024; Biyiri et al., 2025). In many studies, teachers and chatbots jointly curate and adapt learning materials – combining human expertise with instant AI feedback – to foster a collaborative learning dynamic (Adams et al., 2024; Guo et al., 2025).

Chatbots scaffold peer-to-peer feedback by guiding discussion prompts and facilitating collaborative problem-solving tasks, thereby strengthening co-regulation among learners (Al-Hafdi & Alnajdi, 2024). They also catalyse peer influence, as students recommend the tool to classmates and engage in role-play or joint feedback activities mediated by the chatbot (Chen et al., 2020; Rahimi & Mosalli, 2025).

Embedded goal-setting and reflection prompts enable chatbots to orchestrate joint planning and self-evaluation sessions, often with teacher oversight, that reinforce shared SRL practices (Fan et al., 2025; Gazulla et al., 2023). Dynamic, context-sensitive feedback during group tasks further sustains dialogue and mutual reflection (Azevedo et al., 2016).

Beyond mediation, instructors leverage chatbots to scaffold learning – providing just-in-time support and guiding students through complex tasks – while integrating AI tools into curricula to enhance social presence and interactivity (Baillifard et al., 2025; Canonigo, 2024).

Chatbots amplify social presence in online and blended settings by maintaining ongoing dialogues with learners, peers, and teachers, thus fostering a sense of community and belonging (Cai et al., 2025; Huang & Hew, 2025). They also facilitate dialogic learning, prompting reflective conversations that interweave individual and collective SRL processes (Bahari et al., 2025; Mills et al., 2025).

Studies caution that chatbots cannot fully replicate the emotional nuance of human interaction, highlighting the need for richer teacher mediation to preserve interpersonal support (Chang & Sun, 2025). Future research should explore designs that deepen peer-to-peer engagement and extend chatbot roles beyond task facilitation to genuine co-regulatory partners.

4.4 RQ3: how do AI chatbots promote critical self-reflection within SRL, enabling learners to examine and transform their own learning strategies and underlying assumptions?

Our analysis revealed six interlocking mechanisms through which AI chatbots support critical self-reflection in SRL (Table 6). Chatbots guide learners to appraise their own work by highlighting errors, soliciting

Table 5. Social interaction of AI chatbots for SRL.

Category	Subcategory	Specific functionality examples	References
Teacher-Student Interaction	Teacher as a Mediator of Social Interaction	• The role of teachers in guiding the use of AI chatbots to enhance the social dimension of SRL. • Teachers are facilitating AI chatbot interactions, setting up structured activities, and verifying content to enhance collaborative learning. • Teacher-led training sessions and supervised tasks that ensure AI chatbots are used effectively.	(Abdelhalim, 2024) (Chen, 2024) (Chiu, 2024) (Han et al., 2023) (Jeon, 2022) (Kim et al., 2025) (Kong & Yang, 2024) (Kong et al., 2024) (Lee et al., 2022) (Levine et al., 2025) (Naz & Robertson, 2024) (Ng et al., 2024) (Shahzad et al., 2025) (Teng et al., 2025) (Wang, 2024) (Xia, Chiu, & Chai, 2023) (Xiao et al., 2024)
	Teacher-Chatbot Collaboration	• Teachers and AI chatbots working together to curate and verify learning materials, promoting collaborative and interactive student engagement. • The combination of teacher-guided learning with immediate, interactive AI feedback to reinforce social interaction during learning tasks.	(Adams et al., 2024) (Bai et al., 2025) (Guo et al., 2025) (Li, 2023) (Ng et al., 2024) (Saritepeci & Durak, 2024)
Peer Interaction	Peer Feedback and Collaboration	• AI chatbots facilitate peer feedback and collaborative tasks, where students provide corrective feedback to each other. • Chatbots supporting structured peer interactions through guided activities such as peer-to-peer discussions and collaborative problem-solving.	(Al-Hafdi & Alnajdi, 2024) (Darmawansah et al., 2025) (Darvishi et al., 2024) (Fidan & Gencel, 2022) (Guo et al., 2023) (Teng, 2025)
	Peer Influence and Engagement	• The influence of peers on the adoption of AI chatbots, as peers recommend the tool and encourage usage, fostering social learning environments. • Peer-to-peer engagement through AI-mediated tasks like role-playing and collaborative feedback activities.	(Chen et al., 2020) (Chen & Law, 2025) (Li, Wang, et al., 2024) (Rahimi & Mosalli, 2025) (Tai & Chen, 2023) (Tram et al., 2024) (Yin et al., 2024)
Social Learning Strategies	Facilitating Goal Setting and Self-Reflection	• AI chatbots help students set learning goals and reflect on their progress, fostering a social learning environment where teacher and student engage in reflective practices. • Chatbots supporting self-regulation through teacher-mediated reflection prompts and goal-setting activities.	(Fan et al., 2025) (Gazulla et al., 2023) (Han et al., 2022) (Harley et al., 2018) (Huang et al., 2023) (Huang & Lin, 2024) (Huang, 2024) (Jiang et al., 2025) (Jin et al., 2023) (Kaiss et al., 2024) (Khallel, 2024) (Kumar, 2021) (Lee et al., 2022) (Li, Sadiq, et al., 2025) (Liu & Reinders, 2025) (Rezai et al., 2024) (Wang, Lin, et al., 2025) (Wu et al., 2024) (Xia, Chiu, Chai, & Xie, 2023)
	Dynamic Feedback for Collaborative Learning	• AI chatbots provide immediate, personalised feedback that encourages communication between students and the teacher. • The role of AI chatbots in offering contextually relevant prompts that stimulate student engagement and reflection, supported by teacher feedback.	(Guo & Li, 2024) (Robillos, 2024) (Sayed et al., 2024)
Teacher Roles	Instructor Training and Integration	• Teachers' role in training students to use AI tools effectively, ensuring that the use of chatbots supports social interaction and SRL processes. • The integration of AI chatbots into curricula, with the teacher providing context, instructions, and mediation to enhance social interaction.	(Holster, 2024) (Li, 2023) (Liu, Hwang, et al., 2024)
Collaborative and Interactive Learning Environments	Supporting Social Presence	• AI chatbots foster social presence in online learning environments, allowing students to feel more connected to their peers and teachers. • Enhancing student engagement through chatbot-mediated dialogue and group-based interactions, promoting a sense of belonging and community.	(Cai et al., 2025) (Hayashi, 2020) (Huang & Hew, 2025) (Liang et al., 2024) (Xie et al., 2024)
	Interactive and Dialogic Learning	• The role of AI chatbots in facilitating dialogic learning, where students engage in ongoing dialogues with the chatbot, peers, and the teacher. • Using AI to encourage reflective conversations and collaborative discussions that enhance the social aspects of SRL.	(Duong & Chen, 2025) (Durak, 2023) (Han et al., 2025) (Koc-Januchta et al., 2020) (Mills et al., 2025) (Nguyen & Nguyen, 2024) (Pan et al., 2025)
Limitations and Future Directions for Social Interaction with AI Chatbots	Enhancing Peer Interaction through AI	• The potential for AI chatbots to be used more effectively in peer-to-peer interactions, fostering collaborative learning through structured tasks and feedback mechanisms. • The need for	(Chang & Sun, 2025)

(*Continued*)

Table 5. Continued.

Category	Subcategory	Specific functionality examples	References
		further design work to support richer teacher-student and peer-to-peer interactions in AI-mediated environments.	

justification of choices, and encouraging comparison of initial and revised responses – fostering deeper metacognitive awareness of strengths and weaknesses (Baillifard et al., 2025; Darvishi et al., 2024).

Several studies warn against overreliance on automated feedback, arguing that true reflection requires students to contest AI suggestions and integrate their own judgements, thus preserving independent critical thinking (Durak, 2023).

Instructors play a pivotal role by reviewing AI-generated content, prompting students to question its validity, and modelling reflective dialogue – thereby ensuring that chatbot-facilitated reflection remains accurate and contextually grounded (Kim et al., 2025).

Embedded prompts encourage learners to articulate their reasoning, identify knowledge gaps, and plan subsequent strategies; iterative feedback loops prompt repeated cycles of goal-setting, monitoring, and evaluation (Cai et al., 2025).

Chatbots can be designed to surface ethical dilemmas – such as data privacy, algorithmic bias, and the limits of AI authority – to stimulate discussions that expand reflection beyond task performance into broader socio-technical critique (Canonigo, 2024).

Emerging studies advocate for chatbots that empower learners to interrogate their assumptions and systemic power dynamics, transforming self-reflection into a tool for critical consciousness and learner agency (Bahari et al., 2025; Mills et al., 2025).

4.5 Key functional roles of AI chatbots in supporting SRL (RQ4)

The findings synthesised from the technological affordances (RQ1), social interaction dynamics (RQ2), and reflective practices (RQ3) collectively highlight eight distinct but interrelated functional roles through which AI chatbots support SRL (Figure 6). These roles provide an integrated understanding of how chatbot functionalities bridge Habermas's *technical, practical,* and *emancipatory* interests.

First, within the forethought phase, chatbots primarily function as personalised planners and goal-setting facilitators, leveraging adaptive learning algorithms and NLP to provide tailored content, task sequencing, and goal articulation prompts. Additionally, chatbots can serve as motivational facilitators, employing affective computing techniques such as gamified interfaces and adaptive motivational messages to encourage sustained learner engagement.

During the performance phase, the chatbot's roles expand to include real-time task scaffolding and feedback providers, employing diagnostic NLP systems to offer instant corrective feedback and strategy-orientated prompts that assist learners in monitoring and controlling their cognitive processes. Concurrently, chatbots act as social mediators, orchestrating teacher-student-peer interactions, structuring collaborative dialogues, and supporting co-regulated learning through multimodal and interactive environments. Moreover, chatbots function as performance trackers and analytics engines, gathering and visualising interaction data to inform learners and educators about ongoing task progress, strategy effectiveness, and engagement patterns.

In the self-reflection phase, AI chatbots predominantly adopt roles as metacognitive scaffolds, prompting learners to critically evaluate their learning strategies, identify gaps in their understanding, and formulate adaptive revisions. Furthermore, chatbots fulfil the role of ethical-critical reflectors, raising awareness about algorithmic biases and ethical implications of AI guidance and empowering learners to question and critique automated suggestions, thus addressing the emancipatory dimensions of Habermas's framework. Finally, chatbots act as integrative facilitators of reflective dialogue, bridging the individual and collective reflective processes by providing structured opportunities for learners and educators to collaboratively interrogate learning practices and outcomes.

Together, these eight roles constitute a comprehensive typology. This typology clearly illustrates how chatbot functionalities systematically align with and enhance the technical (instrumental efficiency), practical (social coordination), and emancipatory (critical self-awareness) dimensions of SRL.

Table 6. Reflective practices with AI chatbots.

Category	Subcategory	Specific functionality examples	References
Critical Self-Reflection	Self-Evaluation and Critical Reflection	• AI chatbots facilitate self-reflection by prompting learners to critically evaluate their progress and performance. • The role of chatbots in fostering self-assessment, where learners review their responses, mistakes, and growth.	(Chang & Sun, 2025) (Chen et al., 2020) (Chiu, 2024) (Darvishi et al., 2024) (Duong & Chen, 2025) (Fidan & Gencel, 2022) (Guo et al., 2023) (Harley et al., 2018) (Huang & Lin, 2024) (Jeon, 2022) (Kong & Yang, 2024) (Naz & Robertson, 2024) (Nguyen & Nguyen, 2024) (Pan et al., 2025) (Rahimi & Mosalli, 2025) (Robillos, 2024) (Saritepeci & Durak, 2024) (Sayed et al., 2024) (Shahzad et al., 2025) (Tai & Chen, 2023) (Teng et al., 2025) (Tram et al., 2024) (Wu et al., 2024)
	Balancing AI Assistance and Student Agency	• The potential risk of overreliance on AI tools that could undermine students' ability to critically engage with their learning and develop independent thinking. • The need for learners to balance external AI support with their own reflective practices to avoid becoming passive recipients of information.	(Adams et al., 2024) (Durak, 2023) (Fan et al., 2025) (Guo & Li, 2024) (Han et al., 2025) (Huang & Hew, 2025) (Kaiss et al., 2024) (Khallel, 2024) (Kong et al., 2024) (Lee et al., 2022) (Levine et al., 2025) (Li, 2023) (Li, Sadiq, et al., 2025) (Li, Wang, et al., 2024) (Liu, Hwang, et al., 2024) (Liu & Reinders, 2025) (Rezai et al., 2024) (Xia, Chiu, & Chai, 2023) (Xie et al., 2024)
Role of Teachers in Fostering Self-Reflection	Teacher Mediation and Critical Oversight	• Teachers acting as critical mediators in the use of AI chatbots, ensuring that AI outputs are reviewed and adapted to avoid bias or misinformation. • The importance of teacher involvement in guiding students through reflective processes, particularly in verifying and questioning AI-generated content.	(Bai et al., 2025) (Guo et al., 2025) (Kim et al., 2025) (Li, Sadiq, et al., 2025) (Liu, Hwang, et al., 2024) (Naz & Robertson, 2024) (Rezai et al., 2024) (Teng et al., 2025)
	Enhancing Critical Thinking through Teacher-Chatbot Collaboration	• How teachers' engagement with AI chatbots in the classroom can model reflective practices and critical thinking, allowing students to observe and adopt these strategies. • Teachers' role in fostering critical engagement by adapting AI outputs and encouraging students to reflect on these adaptations.	(Al-Hafdi & Alnajdi, 2024)
Limitations of AI Chatbots in Supporting Self-Reflection	Concerns about Misinformation and Accuracy	• The risk of overreliance on AI tools like ChatGPT, which may provide incorrect or superficial information, and the need for critical evaluation from students. • The importance of students developing critical thinking skills to assess the accuracy of AI outputs and incorporate human feedback into their reflective processes.	(Lee et al., 2022) (Lee, 2024) (Naz & Robertson, 2024) (Ng et al., 2024) (Robillos, 2024)
	Potential for Surface-Level Reflection	• The possibility that AI chatbots could foster only superficial reflection, where students perform surface-level evaluations of their work rather than engaging in deeper, critical self-reflection. • The need for instructional designs that encourage deeper, more meaningful reflection, supported by AI, without allowing learners to rely solely on AI-generated feedback.	(Kumar, 2021) (Nguyen & Nguyen, 2024)
AI Chatbots as Tools for Promoting Metacognitive Awareness	Fostering Metacognitive Skills	• AI chatbots promoting metacognitive awareness by prompting students to reflect on their thinking and learning processes. • The role of AI chatbots in helping students recognise gaps in their knowledge and providing opportunities for them to adjust their learning strategies.	(Abdelhalim, 2024) (Cai et al., 2025) (Chen, 2024) (Chen & Law, 2025) (Darmawansah et al., 2025) (Gazulla et al., 2023) (Hayashi, 2020) (Huang et al., 2023) (Koc-Januchta et al., 2020) (Liang et al., 2024) (Pan et al., 2025) (Saritepeci & Durak, 2024) (Teng, 2025) (Wang, Zhong, et al., 2024) (Xiao et al., 2024) (Yin et al., 2024)
	Encouraging Iterative Self-Reflection	• The iterative nature of self-reflection supported by AI chatbots, where students continuously engage in reflective feedback loops that help refine their	(Huang, 2024) (Pan et al., 2025)

(*Continued*)

Table 6. Continued.

Category	Subcategory	Specific functionality examples	References
		learning strategies. • How chatbots guide students through repetitive tasks that prompt critical thinking and reflection, promoting a cycle of self-improvement.	
Ethical and Critical Engagement with AI Chatbots in Self-Reflection	Ethical Use of AI and Reflective Practices	• The need for students and educators to engage critically with the ethical implications of AI chatbot use, including issues of data privacy, bias, and the potential dehumanisation of learning processes. • How AI chatbots can be designed to foster critical ethical reflection, ensuring that students question the validity and fairness of AI-generated content.	(Holster, 2024) (Rahimi & Mosalli, 2025) (Rezai et al., 2024) (Sayed et al., 2024) (Shahzad et al., 2025) (Tai & Chen, 2023) (Teng et al., 2025)
	Addressing the Risk of Over-reliance and Diminished Reflection	• The risk that over-reliance on AI chatbots could diminish students' ability to engage in genuine self-reflection and critical evaluation of their own learning. • The need for careful design of AI interventions that encourage reflection without fostering dependence on AI-generated content or responses.	(Liu, Hwang, et al., 2024) (Rahimi & Mosalli, 2025) (Robillos, 2024) (Teng et al., 2025) (Wang, 2024) (Wang, Lin, et al., 2025)
The Future of AI Chatbots in Supporting Emancipatory Reflection in SRL	Designing for Empowerment and Reflection	• How future designs of AI chatbots can incorporate features that foster deeper self-reflection and empowerment, encouraging learners to critically engage with their learning and its broader implications. • The role of AI in facilitating emancipatory education by supporting learners in questioning their assumptions and engaging in critical, transformative reflection.	(Bahari et al., 2025) (Han et al., 2022) (Han et al., 2023) (Jiang et al., 2025) (Jin et al., 2023) (Liu & Reinders, 2025) (Mills et al., 2025) (Naz & Robertson, 2024) (Nguyen & Nguyen, 2024) (Saritepeci & Durak, 2024) (Sayed et al., 2024) (Shahzad et al., 2025) (Tai & Chen, 2023) (Xia, Chiu, & Chai, 2023)

5. Discussion

This study affords two theoretical contributions (see sections 5.1 and 5.2) and three practical suggestions (see sections 5.3, 5.4 and 5.5).

5.1. A habermasian perspective on chatbot-mediated SRL

This Habermas's framework maps the chatbot functions directly onto Zimmerman's SRL phases and Habermas's three interests, providing a novel and holistic model for understanding, designing, and evaluating AI chatbots for SRL (Figure 7). This moves beyond descriptive cataloguing to offer an explanatory model for how and why different chatbot functionalities support learning. Specifically, chatbot-generated metacognitive prompts and adaptive analytics embody the technical interest by efficiently facilitating learners' strategic monitoring, self-assessment, and feedback processes. These technologically driven functionalities streamline SRL and optimise learners' ability to engage with complex learning tasks effectively (Chen et al., 2025; Labadze et al., 2023).

Simultaneously, teacher- and peer-mediated interactions facilitated by chatbots emphasise Habermas's practical (communicative) interest. Structured collaborative dialogues, peer-to-peer feedback cycles, and teacher-guided reflective sessions significantly enhance learners' shared understanding and co-regulation skills. The social presence fostered through these chatbot-mediated interactions creates authentic opportunities for learners to negotiate meaning, articulate strategies, and mutually reinforce metacognitive awareness, thereby extending regulation from an individual cognitive act into a socially situated process (Chen, 2024; Huang & Hew, 2025).

Critically, chatbot prompts designed to stimulate deeper reflection on learning processes, assumptions, and ethical considerations align with Habermas's emancipatory interest, fostering learner agency and transformative insight. These prompts encourage learners to critically evaluate the validity and potential biases inherent in AI-generated guidance, promoting reflective and critical consciousness as central outcomes of the SRL process (Cai et al., 2025; Mills et al., 2025). However, the empirical evidence presents both



Figure 6. Roles of AI chatbots in SRL.

promise and caution. While several studies highlight substantial improvements in learners' reflective depth, metacognitive accuracy, and strategy adaptation following sustained chatbot-supported reflection, others report predominantly superficial or descriptive reflection that lacks analytical depth (Durak, 2023). This variability underscores the necessity of pedagogically robust chatbot design and intentional instructional mediation to cultivate meaningful reflective practices.

So, the chatbot becomes an important tool that connects practical use, social conversation, and thoughtful reflection in a smooth and active learning process. Future research should specifically look at how these three Habermasian dimensions interact in various educational settings and explore the conditions that allow chatbot-mediated SRL to have the greatest transformative effect.

5.2 Methodological analysis

Our analysis provided critical insights into the strengths and limitations of current research on AI chatbot-supported SRL. Predominantly, studies employed mixed-methods approaches, leveraging self-report measures such as surveys and inventories to assess learner perceptions, strategy use, and motivational outcomes. While self-reports offer valuable insights into learners' subjective experiences, an overreliance on these methods risks introducing response biases and limiting the depth of understanding about the actual learning processes and behaviours (Ifenthaler et al., 2023). Moreover, short-term studies dominated literature, restricting our understanding of the long-term impacts and sustainability of chatbot-facilitated SRL.

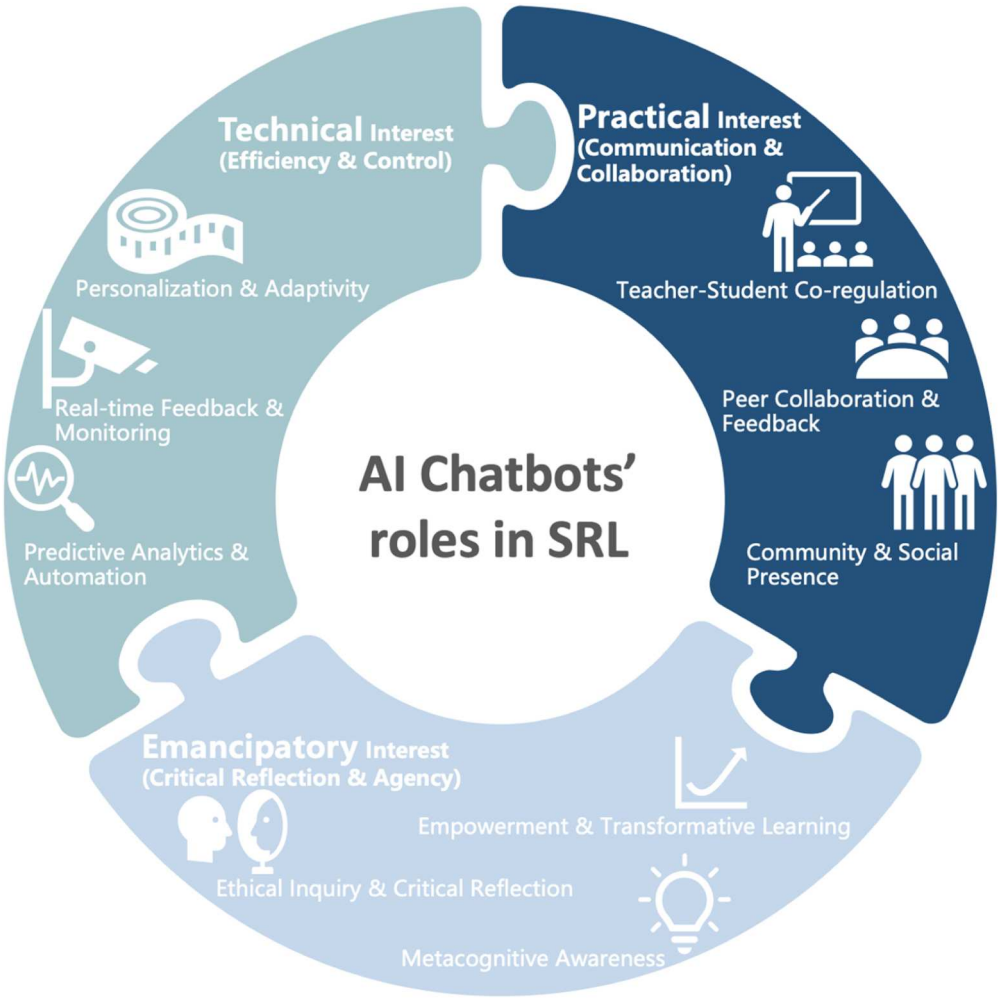


Figure 7. AI chatbots' roles in SRL based on Habermas' Knowledge-Constitutive Interests.

To address these limitations, future studies should prioritise longitudinal designs that can trace the evolving nature of SRL processes over extended periods. Such designs are essential to understanding how initial chatbot interactions influence sustained learner autonomy, motivation, and reflection over weeks, months, or even semesters. Furthermore, there is a pressing need for integrating richer and more varied multimodal data collection methods, such as ecological momentary assessments, physiological sensors, eye-tracking, and real-time learning analytics (Olsen et al., 2020; Ronkainen et al., 2025). These advanced methodological tools can provide fine-grained, real-time insights into learners' cognitive, emotional, and metacognitive dynamics, thus offering a more nuanced picture of SRL processes as they unfold within authentic learning contexts.

Additionally, our review revealed a strong concentration of research within higher education contexts, with limited attention to primary and secondary education settings or professional development contexts. Future empirical studies should systematically investigate diverse educational contexts and learner populations, including K-12 settings, adult learning environments, and vocational training. Such diversity is crucial for enhancing the external validity and generalisability of findings and for identifying context-specific affordances and constraints in chatbot designs. Finally, incorporating participatory research methods – engaging educators and learners directly in the design, implementation, and evaluation processes – would further enrich methodological rigour and relevance (Stanton, 2025), ensuring that chatbot interventions are culturally responsive, pedagogically meaningful, and aligned with the diverse needs of learners.

5.3 Technical interest: enhancing efficiency and optimisation

Drawing on Habermas's technical interest, concerned with controlling processes to achieve greater efficiency and optimisation (Habermas, 2015), our review demonstrates that AI chatbots furnish powerful affordances for personalised, data-driven SRL support. In particular, chatbots tailor learning trajectories by adjusting task difficulty and sequencing in real time, delivering context-sensitive feedback, and scaffolding goal setting (Adams et al., 2024; Bahari et al., 2025). Such dynamic adaptation not only sustains learner engagement but also streamlines the cognitive and metacognitive operations that underpin Zimmerman's SRL cycle. Moreover, predictive analytics embedded within chatbot platforms can model learner behaviour to predict performance risks and suggest timely interventions (Teng et al., 2025). Automated scoring and progress dashboards enable learners to allocate resources to strategic planning and reflection (Darmawansah et al., 2025; Sayed et al., 2024). Nevertheless, privileging efficiency carries the risk of rendering learning interactions overly transactional. Without careful pedagogical design, learners may hand over metacognitive control to the system, thereby weakening opportunities for self-initiated reflection and strategic decision-making (Durak, 2023). Future design must therefore embed desirable difficulties within chatbot workflows – such as open-ended prompts or delayed feedback – to preserve space for deeper cognitive engagement and prevent automation complacency.

5.4 Practical interest: fostering communication and collaboration

Under Habermas's practical interest – aimed at enhancing communicative action and mutual understanding (Habermas, 2015) – AI chatbots emerge as mediating agents that enrich the social dimensions of SRL. Our synthesis indicates that chatbots facilitate peer-to-peer interaction through structured dialogues, co-construction tasks, and scaffolded feedback cycles (Fidan & Gencel, 2022; Guo et al., 2023). For example, chatbots can orchestrate jigsaw activities in which learners alternate between information-seeking with the bot and knowledge-sharing with peers, thereby reinforcing both individual and collective regulation processes (Chen, 2024). By nurturing social presence and belonging – particularly in online or blended contexts – chatbots help mitigate isolation and sustain motivation (Xiao et al., 2024). However, the communicative capacity of AI remains bounded: current natural language models cannot fully replicate the pragmatics and empathy of humans (Ng et al., 2024). To address this, designers should integrate multimodal cues (e.g. prosody, sentiment analysis) and enable teacher oversight in chatbot-mediated discussions, ensuring that critical misunderstandings are surfaced and that discourse remains pedagogically purposeful (Cai et al., 2025).

5.5 Emancipatory interest: promoting critical reflection and learner agency

Habermas's emancipatory interest seeks to transform power relations and foster autonomy through critical reflection (Habermas, 2015). In our review, AI chatbots supported emancipatory SRL by prompting learners to articulate their reasoning, question assumptions, and evaluate strategy effectiveness (Baillifard et al., 2025; Pan et al., 2025). These reflective dialogues cultivate metacognitive awareness and empower students to take ownership of their learning trajectories. Furthermore, by exposing learners to scenario-based ethical dilemmas – such as algorithmic bias or data privacy trade-offs – chatbots can catalyse critical discourse on the societal dimensions of AI (Mills et al., 2025; Pleasants & Radloff, 2024; Tai & Chen, 2023).

Yet, emancipatory potential is not intrinsic to the technology; it depends on the interplay between automated prompts and human facilitation. Without instructor-designed reflection scaffolds, chatbot questions may remain superficial, yielding limited transformative insight (Durak, 2023; Robillos, 2024). Future work should investigate hybrid designs in which educators co-design critical reflection modules with AI, thereby ensuring that emancipatory prompts are culturally responsive and aligned with disciplinary epistemologies.

5.6 Limitations and future research

We acknowledge four limitations in this study. First, an overemphasis on automation can inadvertently undermine learner agency: when chatbots assume too much control over pacing, feedback, or goal-

setting, students may become passive recipients rather than active regulators of their learning. This is a concern that has been documented in teacher populations. Second, while AI-driven conversations can help students interact with each other, today's chatbot designs still have trouble understanding the subtleties, feelings, and practical aspects of human communication, so teachers need to be involved to clear up any confusion and keep discussions meaningful. This requires teachers to understand what chatbots are, their affordances and limitations, and their ethical implications (Heath et al., 2022). Third, while chatbots can encourage thinking, how deep and impactful that thinking is relies a lot on careful teaching design; without well-planned support, the prompts for reflection might not go very deep. To fill these gaps, future research should look into "calibrated autonomy" models that change the mix of automated help and learner control, study chatbot interfaces that use emotional and social cues to improve communication, and involve teachers and students together in creating meaningful reflection activities. By tackling these areas, the field can move beyond viewing AI chatbots merely as efficient tools and towards realising their full potential as catalysts for deeply engaged, socially situated, and critically reflective SRL. Last, this study focuses on Scopus and Web of Science, may miss some articles published in other databases. Future studies should include more databases such as PsycINFO, ERIC, and ProQuest.

Authors' contributions

Xiuyi WU: Conceptualisation, Methodology, Formal analysis, Validation, Writing - Original Draft, Writing - Review & Editing; **Jeffrey D. RADLOFF:** Writing - Review & Editing; **Ibrahim H. YETER:** Writing - Review & Editing; **Wei WANG:** Writing - Review & Editing; **Thomas K.F. CHIU:** Writing - Review & Editing, Supervision, Funding acquisition.

Consent for publication

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The datasets used for the current study are available from the corresponding author on reasonable request.

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